THARUNKUMAR K

SOFTWARE DEVELOPMENT ENGINEER | FULL STACK DEVELOPER | AI SYSTEMS

Dharapuram, Tamil Nadu, India | tharunkumark42007@gmail.com | +91 87609 64830

github.com/Tharun4743 | linkedin.com/in/tharunkumark4743 | leetcode.com/u/Tharunkumar__K | tharunkumark4743.netlify.app

PROFESSIONAL SUMMARY

Information Technology undergraduate (CGPA: 8.5/10) with practical experience in full-stack engineering, Retrieval-Augmented Generation (RAG), and real-time collaborative applications. Proficient in Python, Java, TypeScript, Node.js, and PostgreSQL, with experience building offline AI pipelines, sandboxed execution runtimes, and campus platforms. Strong foundations in Data Structures, Algorithms, Object-Oriented Programming, and database systems.

TECHNICAL SKILLS

Programming Languages:	Python, Java, C, C++, TypeScript, JavaScript, SQL
AI & Machine Learning:	Retrieval-Augmented Generation (RAG), ChromaDB, Gemini API, Ollama (LLaMA 3), Prompt Engineering
Backend & Web:	Node.js, Express.js, Spring Boot, REST APIs, WebSockets, Socket.IO, WebRTC, React, Tailwind CSS
Databases & Storage:	PostgreSQL, Supabase (with RLS), SQLite, MongoDB, ChromaDB, Cloudinary
Developer Tools:	Git, GitHub, GitHub Actions, Linux / Bash, Postman, Vite, Electron, Vercel, Render
Core Computer Science:	Data Structures & Algorithms (DSA), Object-Oriented Programming (OOP), DBMS, Concurrency

EDUCATION

Bachelor of Technology (B.Tech) in Information Technology | CGPA: 8.5 / 10.0 **Batch 2024 – 2028**

VSB Engineering College, Karur, Tamil Nadu **Expected May 2028**

PROFESSIONAL EXPERIENCE

Fullstack Development Intern — Neura Global **06/2026 – 07/2026**

- **Campus Placement Portal (CampusConnect):** Built core modules for a placement management portal; implemented 4-tier Role-Based Access Control (Admin, Recruiter, Student, Coordinator) and automated eligibility filtering.
- **Real-Time Pipelines & Document Storage:** Integrated Socket.IO WebSocket pipelines for interview status updates; connected Cloudinary for candidate document uploads with CSRF protection to streamline recruitment workflows.

Software Development Intern — Infosys Springboard **11/2025 – 01/2026**

- **Employee Recognition Platform:** Completed the Infosys Springboard 6.0 program; built the **StarWall** web application for employee peer recognition, milestone tracking, and organizational analytics.
- **Collaborative Engineering Discipline:** Practiced production Git workflows, modular code architecture, peer code reviews, and structured debugging in a collaborative team setup.

FEATURED PROJECTS

VSBEC IT Vault — Academic Task Manager & Code Evaluation Platform | *Top 50 SIH Nominee* **2026**

React 19, TypeScript, Tailwind CSS, Node.js, Express, PostgreSQL, Monaco Editor [Demo] | [GitHub]

- **Code Execution Sandbox:** Built a multi-language code evaluation runtime (C, C++, Java, Python) using isolated child processes with 4-second timeout traps, memory boundaries, and automated evaluation against hidden test cases.
- **Campus Scale & Performance:** Deployed an academic task management platform actively used daily by **365+ enrolled students**, designed with 35 relational PostgreSQL tables, role-based access control, and fast in-memory caching.

GOAT Code Editor — Real-Time Collaborative Web IDE | *1st Place National Winner* **2026**

React 19, TypeScript, WebRTC, Socket.IO, Monaco Editor, Express.js, PostgreSQL, Piston API [Demo] | [GitHub]

- **Real-Time Collaboration & WebRTC:** Built a collaborative web code editor with live syntax diffing and cursor presence using Monaco Editor and Socket.IO; integrated peer-to-peer WebRTC audio communication mesh.
- **Code Execution & Resilience:** Integrated Piston API for in-browser execution across 13+ languages; implemented dual-persistence pairing PostgreSQL with an in-memory Map fallback. Won 1st Place at Code Thugs 2K26 (INR 5,000 cash prize).

AURA — AI Unified Retrieval Assistant (100% Offline Multimodal RAG) **2026**

Java 21, Spring Boot, React 19, Vite, SQLite, ChromaDB, Ollama (LLaMA 3), Electron [GitHub]

- **Offline Retrieval-Augmented Generation:** Developed an offline RAG desktop application for SIH PS-25231; extracted PDF text with Apache PDFBox and indexed embeddings in ChromaDB for local semantic retrieval without cloud egress.
- **Local Inference & Streaming:** Connected local LLaMA-3 models running via Ollama over local loopback; implemented WebSocket token streaming for real-time responses and stored query logs locally in SQLite.

Smart Helmet IoT — Rider Safety & Accident Prevention System | *Top 50 SIH Nominee* **2025**

Embedded C/C++, Arduino Nano, Arduino Uno, RF 433MHz Telemetry, MQ-3 Sensor, Relay [GitHub]

- **Dual-Unit Telemetry & Failsafe:** Engineered wireless 433MHz telemetry transmitting 100ms serialized packets with checksums; built a 5-second signal-loss watchdog that automatically engages ignition interlock upon tamper or loss.
- **Proactive Safety & Sub-500ms Interlock:** Interfaced MQ-3 alcohol sensor and optical IR wear sensor to enforce sobriety and helmet compliance; achieved sub-500ms ignition cutoff; shortlisted in Top 50 of SIH 2025.

HONORS, ACHIEVEMENTS & LEADERSHIP

- **1st Place National Winner** — *Code Thugs 2K26* (National Hackathon Champion for GOAT Collaborative IDE; INR 5,000 cash prize).
- **3rd Prize Winner** — *Neural Minds 8-Hour AI Hackathon 2026* (Awarded by Dept of AI&ML for FINDORA AI multimodal system).
- **Top 50 / 300+ Teams (2x Nominee)** — *Smart India Hackathon (SIH 2026 & SIH 2025)* (VSBEC IT Vault & Smart Helmet IoT).
- **National Finalist (2x)** — *Odoo x NMIT Bangalore Hackathon 2026* (Work Suite HRMS) & *India Innovates 2026* (Public CRM).
- **Department Student Coordinator** — *VSB Engineering College* (Leading student technical activities, workshops, and hackathons).

VERIFIED CERTIFICATIONS

AI & Machine Learning:	Google Cloud: Responsible AI, Prompt Design in Vertex AI, Intro to LLMs; Tata GenAI
Software & Systems:	Infosys Springboard Software Development, Cisco Intro to IoT, TCS iON Career Edge, Pega 60-Hour NIP